👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3C

Ratio Reasoning: Convert Measurements within the Same System

Lesson 3-6 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use ratio reasoning to convert between units within the same measurement system.

Language Objective: I can explain a conversion using the words unit ratio, convert, measurement system, and per.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Conversion factor · **Unit ratio** · **Convert** · **Measurement system** · **Double number line** · **Ratio table** · **Customary units** · **Metric units**

Level 3: try the blanks from memory first, then check the bank.

1 The ratio that trades one unit for another is a _____ _____.

2. A ratio comparing an amount to exactly _____ unit is a unit ratio.

3. To change a measurement to another unit without changing the amount is to _____.

4. Inches and feet belong to the customary _____; centimeters and meters to the metric one.

5. Two lines whose matching ticks show the same amount in two units form a _____ _____ _____.

6. A table whose every column holds the same comparison is a _____ _____.

7. Inches, feet, pounds, and gallons are _____.

8. Meters, grams, and liters are _____.

Write About the Math

How many cup-scoops of broth does the recipe need, and how do you know?
¿Cuántas tazas de caldo necesita la receta y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Conversion factor** *Factor de conversión*
- ☐ **Unit ratio** *Razón unitaria*
- ☐ **Convert** *Convertir*

- [ ] **Measurement system**
  Sistema de medidas
- [ ] **Double number line**
  Recta numérica doble

Model: *Find the scale factor (120 divided by 8 = 15), then multiply each ingredient by it.* — Students find the scale factor of 15 and explain that multiplying both ingredients by 15 keeps the recipe proportional (5 cups tomatoes becomes 75, 3 cups mozzarella becomes 45).

Start

Complete the frames. Use the word bank.
Completa las oraciones. Usa el banco de palabras.

- **The unit ratio is ___ cups to 1 quart. Cups are ___ than quarts, so I ___ to convert. 3 × 4 = ___ cups. The amount of broth did not change — only the ___ did.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.
Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Conversion factor
2. **Notes 2:** Unit ratio
3. **Notes 3:** Convert
4. **Notes 4:** Measurement system
5. **Notes 5:** Double number line
6. **Notes 6:** Ratio table
7. **Notes 7:** Customary units

8. **Notes 8:** Metric units
9. **Watch for this mistake:** *A common mistake when converting within a measurement system is choosing the operation by habit instead of by unit size — multiplying every time. Converting 36 inches to feet by multiplying gives 36 × 12 = 432 feet, taller than a 40-story building. Ask first which unit is smaller: it takes MORE small units to make the same amount, so going to a smaller unit multiplies, and going to a larger unit divides.*
10. **Try It 1:** — *Every unit ratio says how many of the smaller unit fit inside exactly 1 of the larger one.*
11. **Try It 2:** — *Conversions in this lesson stay inside one system — customary with customary, metric with metric.*